

M1S5 | Software Design Principles

General principles

DRY (Don't Repeat Yourself)

accent your development on avoiding code duplication

KISS (Keep It Simple, Stupid)

system should be as simple as possible

YAGNI (You Aren't Gonna Need It)

avoid useless functionality, focus on current tasks

SOLID 🦴

SLAP (Single Level of Abstraction Principle)

Main Idea

each method should operate the data of one abstraction level

Code as a story

project as the instructions pipeline

Readability

clear and easy-readable code

Simple Testing

you can write tests for each function as for project unit

API is a...

a set of rules, protocols, and tools that
allow different programs to interact with
each other

APIs

REST

most popular one

RPC

remote methods calling as local ones

GraphQL

Facebook's REST analogue

WebSocket

stateful keep-alive client-server
connection

RESTful API

is an architectural style used to create distributed systems.
It defines a set of principles that govern interactions between clients and servers

REST Principles

Client-Server

clients send requests to the server, and the server processes these requests and responds to them

Stateless

server does not store client state information between requests

Caching

clients can cache server responses and reuse them if the requests are identical to previous ones

Unified Interface

unified data transfer methods and formats for client-server interaction

Individual Resources with ID

each resource must have a unique identifier through which it can be identified and accessed for request processing

Resources Representation

clients can choose the appropriate data representation format in the request